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MID TERM EXAMINATION
B.TECH PROGRAMMES (SECOND SEMESTER)
Second Semester, April 2024

Paper Code: BS-112**Subject: Applied Mathematics-II****Time: 1½ Hrs.****Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1. (a)	Convert the complex number $-16 / (1 + i\sqrt{3})$ into Polar form.	2.5	1
(b)	Use Cauchy Integral Formula to evaluate: $\int_C [(\sin \pi z^2 + \cos \pi z^2) / ((z - 1)(z - 2))] dz$, where C is $ z = 3$.	2.5	1
(c)	Find Invariant Points in $w = (3z - 4) / (z - 1)$.	2.5	2
(d)	Determine the residue of function at each Pole, if $f(z) = z^2 / [(z - 1)(z - 2)^2]$.	2.5	2
2. (a)	Find the Analytic function $f(z) = u + iv$, given that $v = e^x (x \sin y + y \cos y)$.	5	1
(b)	Evaluate the Integral $I = \int_C \operatorname{Re}(z^2) dz$ from $z = 0$ to $2 + 4i$ along the Parabola $y = x^2$.	5	1
3. (a)	Show that $\int_0^{2\pi} [d\theta / (a + b \cos \theta)] = 2\pi / (a^2 - b^2)^{1/2}$, for $a > b > 0$, using Contour Integration.	5	2
(b)	Find the Laurent's expansion of $f(z) = 1 / [z(z^2 - 3z + 2)]$ for the region $1 \leq z < 2$.	5	2
4. (a)	Find the image of circle $x^2 + y^2 = 4y$ under the transformation $w = 1/z$.	5	2
(b)	Discuss the analyticity of the function $f(z) = z \bar{z}$ at the origin.	5	1

**** End of Question Paper ****